**Cambridge International Examinations**
Cambridge International General Certificate of Secondary Education

**CHEMISTRY** **0620/42**

Paper 4 Theory (Extended) **March 2017**

MARK SCHEME

Maximum Mark: 80

**Published**

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| Question | Answer | Marks |
|---|---|---|
| 1(a)(i) | **A** | **1** |
| 1(a)(ii) | **B and C** | **1** |
| 1(a)(iii) | **D** | **1** |
| 1(a)(iv) | **A** | **1** |
| 1(a)(v) | **A** | **1** |
| 1(a)(vi) | carbon dioxide **and** water | **1** |
| 1(b)(i) | bitumen | **1** |
| 1(b)(ii) | refinery gas | **1** |
| 1(b)(iii) | refinery gas | **1** |
| 1(b)(iv) | kerosene | **1** |
| 1(b)(v) | heated / boiled / evaporated / vaporised | **1** |

| Question | Answer | Marks |
|---|---|---|
| 2(a) | **M1** filter | **1** |
| | **M2** wash (the residue) using water | **1** |
| | **M3** dry the residue between filter papers / in a warm place | **1** |
| 2(b)(i) | $2^-$ | **1** |
| 2(b)(ii) | $2Ag^+ + Cr_2O_7^{2-} \rightarrow Ag_2Cr_2O_7$ | **1** |
| 2(c) | **M1** Universal Indicator turns blue | **1** |
| | **M2** <u>ammonia / $NH_3$</u> (is made) | **1** |
| 2(d)(i) | **M1** dichromate ions / particles are heavier (than silver ions) | **1** |
| | **M2** so dichromate ions diffuse / move more slowly **ORA** | **1** |
| | **M3** (where they meet they react and) silver dichromate is made | **1** |
| 2(d)(ii) | **M1** red solid forms in less than five minutes **or** red solid forms faster / sooner | **1** |
| | **M2** particles / ions move faster | **1** |
| 2(e)(i) | **M1** breaking down | **1** |
| | **M2** when heated | **1** |


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| Question | Answer | Marks |
|---|---|---|
| 2(e)(ii) | **M1** formula of chromium(III) oxide | **1** |
| | **M2** rest of equation correct to give a fully correct equation ($(NH_4)_2Cr_2O_7 \rightarrow N_2 + Cr_2O_3 + 4H_2O$ scores [2]) | **1** |

| Question | Answer | Marks |
|---|---|---|
| 3(a) | any 2 from:<br>• forward and backward reactions occur at equal rates<br>• amounts / moles / concentrations (of substances) remain constant<br>• closed system | **2** |
| 3(b) | **M1** (particles) have more energy **OR** (particles) move faster | **1** |
| | **M2** more collisions per second **OR** greater collision rate | **1** |
| | **M3** more (of the) particles / collisions have energy greater than the activation energy **OR** more particles / collisions have sufficient energy to react **OR** a greater percentage / proportion / fraction of collisions are successful | **1** |
| 3(c) | **M1** equilibrium moves left / yield decreases | **1** |
| | **M2** because the forward reaction is exothermic **OR** because the reverse reaction is endothermic | **1** |
| 3(d) | **M1** no change | **1** |
| | **M2** numbers of moles of gas on each side is the same | **1** |
| 3(e)(i) | **M1** all bonding pairs correct (two pair of electrons shared between O and N **AND** one pair of electrons shared between N and C*l*) | **1** |
| | **M2** four non-bonding electrons on O **AND** two non-bonding electrons on N **AND** six non-bonding electrons on C*l* to give a fully correct diagram | **1** |
| 3(e)(ii) | **M1** weak forces (of attraction) | **1** |
| | **M2** between molecules / intermolecular | **1** |


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| Question | Answer | Marks |
|---|---|---|
| 4(a)(i) | arrow labelled **A** on or near wire going in an anti-clockwise direction | **1** |
| 4(a)(ii) | arrow labelled **B** in electrolyte pointing towards the cathode | **1** |
| 4(b)(i) | electrons are lost | **1** |
| 4(b)(ii) | **M1** $Cu^{2+}$ ions on left | **1** |
| | **M2** rest of equation correct and correctly balanced ($Cu^{2+} + 2e^- \rightarrow Cu$ scores [2]) | **1** |
| 4(c) | **M1** anode mass decreases | **1** |
| | **M2** copper lost as <u>ions</u> **OR** copper (atoms) becomes <u>ions</u> **OR** $Cu \rightarrow Cu^{2+} + 2e^-$ | **1** |
| | **M3** cathode mass increases | **1** |
| | **M4** copper deposited / layer of copper forms / copper collected at cathode **OR** $Cu^{2+} + 2e^- \rightarrow Cu$ | **1** |

| Question | Answer | Marks |
|---|---|---|
| 5(a) | carbon dioxide <u>reacts</u> with carbon / coke **OR** $CO_2 + C \rightarrow 2CO$ | **1** |
| 5(b) | **M1** $CaO + SiO_2 \rightarrow CaSiO_3$ | **1** |
| | **M2** CaO is a base | **1** |
| | **M3** $SiO_2$ is an acid | **1** |
| 5(c)(i) | (the carbon makes the iron too) brittle | **1** |
| 5(c)(ii) | reacted with oxygen / oxygen blown in | **1** |
| 5(d)(i) | zinc blende | **1** |
| 5(d)(ii) | $2ZnS + 3O_2 \rightarrow 2ZnO + 2SO_2$ | **1** |
| 5(d)(iii) | any 2 from:<br>• forms acid rain<br>• kills trees / plants<br>• kills fish<br>• damages (limestone / marble) buildings / statues<br>• causes breathing difficulties | **2** |
| 5(d)(iv) | **M1** zinc boils | **1** |
| | **M2** (both) impurities do not boil because their boiling point is above 1000 °C | **1** |


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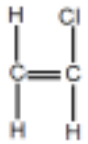
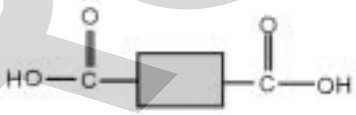
| Question | Answer | Marks |
|---|---|---|
| 6(a)(i) | **M1** (relative formula mass $BaCO_3$ =) 197 | **1** |
| | **M2** (10.0/197 = ) 0.0508<br>(0.0508 alone scores [2]) | **1** |
| 6(a)(ii) | 1.22 | **1** |
| 6(b) | 2.24 | **1** |
| 6(c)(i) | 0.00219 | **1** |
| 6(c)(ii) | **M1** moles HC*l* = 2 × 0.00219 **OR** correct evaluation of this (= 0.00438) | **1** |
| | **M2** (0.00438/0.01875) = 0.234<br>(0.234 alone scores [2]) | **1** |

| Question | Answer | Marks |
|---|---|---|
| 7(a)(i) | circle drawn round two consecutive carbons which includes 3 H atoms and 1 C*l* atom | **1** |
| 7(a)(ii) |  | **1** |
| 7(a)(iii) | **M1** stays yellow / orange / brown or no change | **1** |
| | **M2** becomes colourless | **1** |
| 7(b)(i) | polyamide | **1** |
| 7(b)(ii) | circle must include exactly two C=O, two N–H, one shaded square and one unshaded square | **1** |
| 7(b)(iii) | **M1**<br>  | **1** |
| | **M2**<br> | **1** |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| Question | Answer | Marks |
|---|---|---|
| 7(c)(i) | **M1**<br>34.61/12 : 61.54/16 : 3.85/1 **OR** 2.885 : 3.846 : 3.85 | **1** |
| | **M2**<br>2.885/2.885 : 3.846/2.885 : 3.85/2.885 **OR** 1 : 1.3(33) : 1.3(33) **OR** 3:4:4 | **1** |
| | **M3**<br>$C_3O_4H_4$ | **1** |
| 7(c)(ii) | relative formula mass / relative molecular mass | **1** |

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